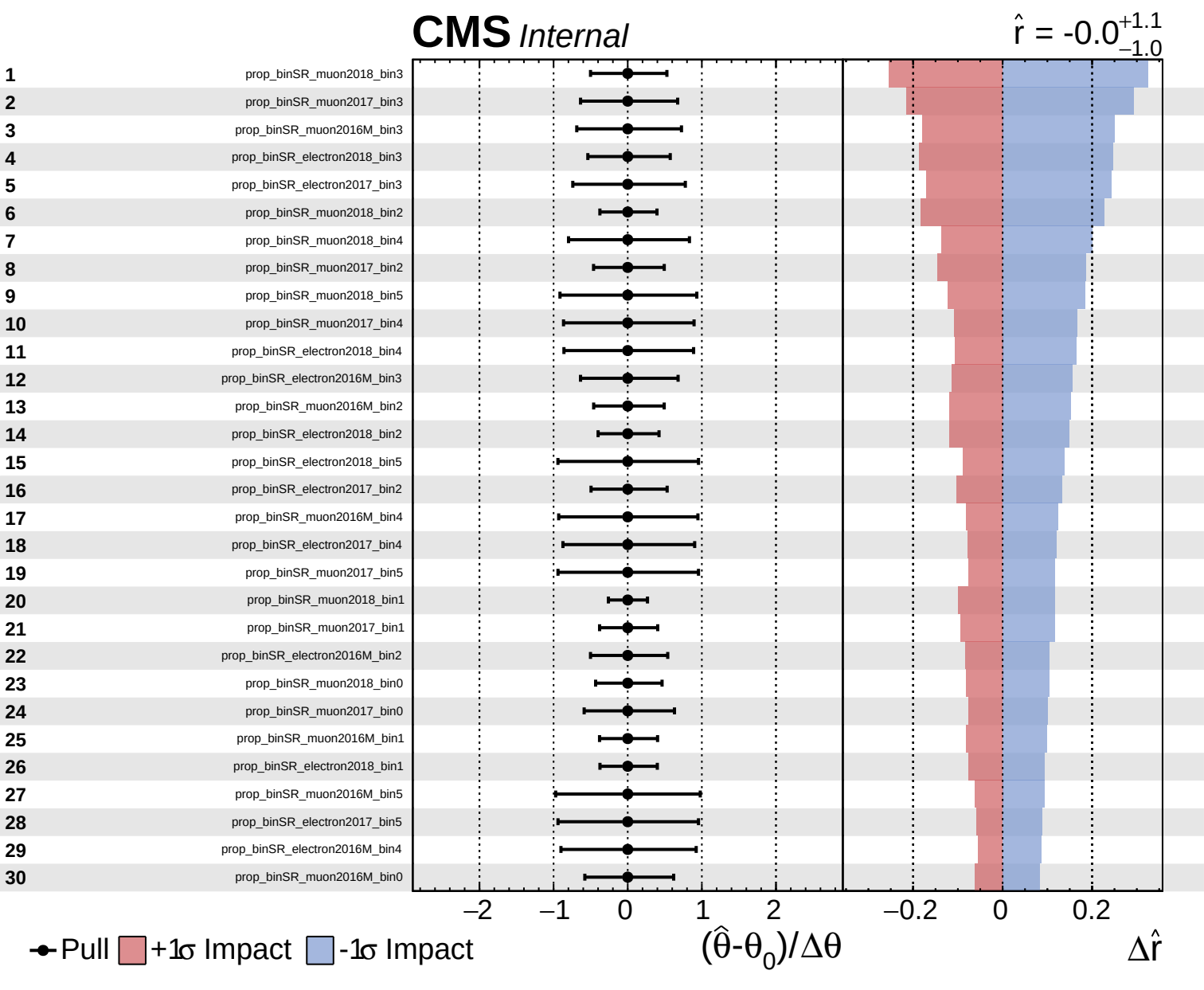
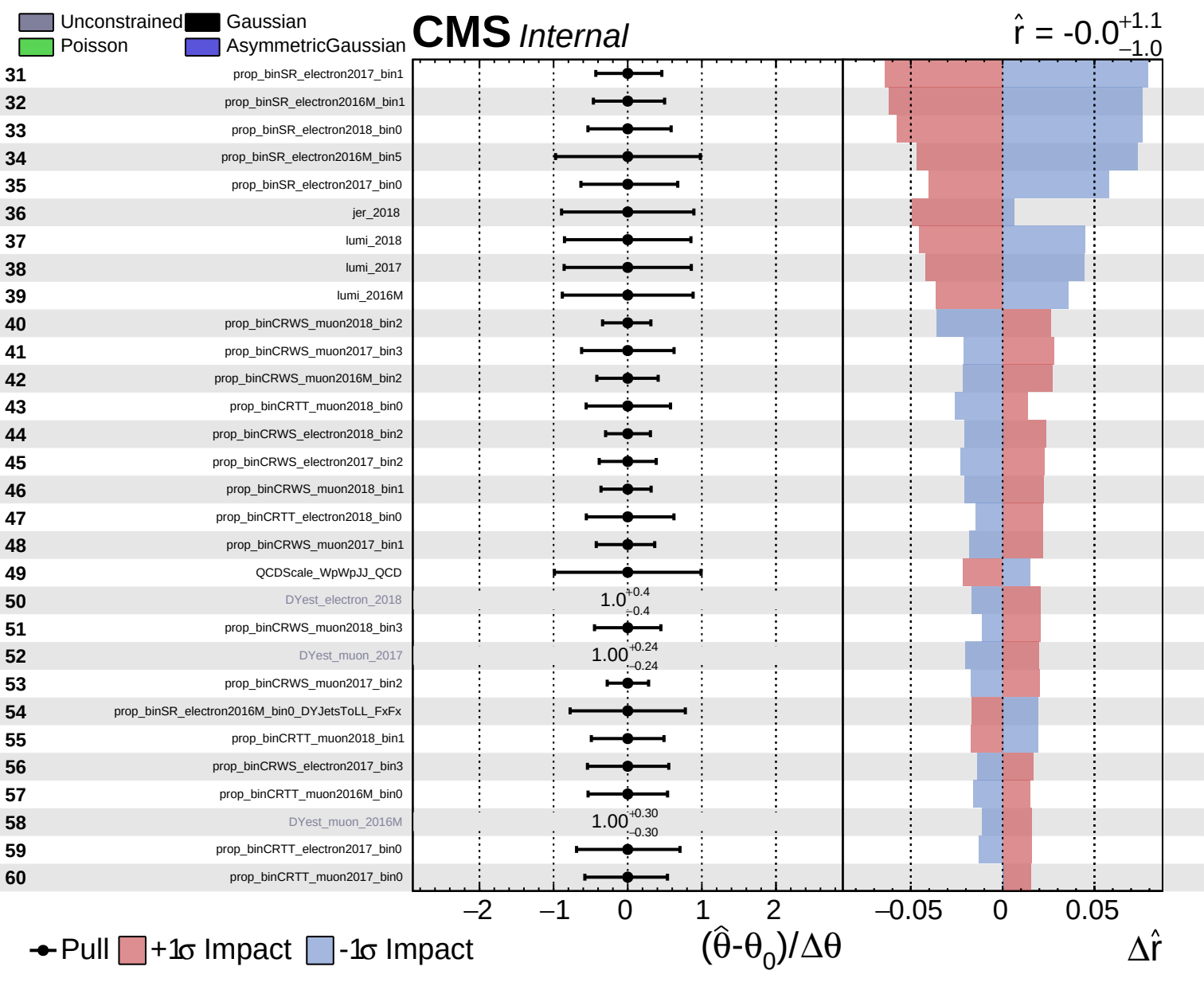
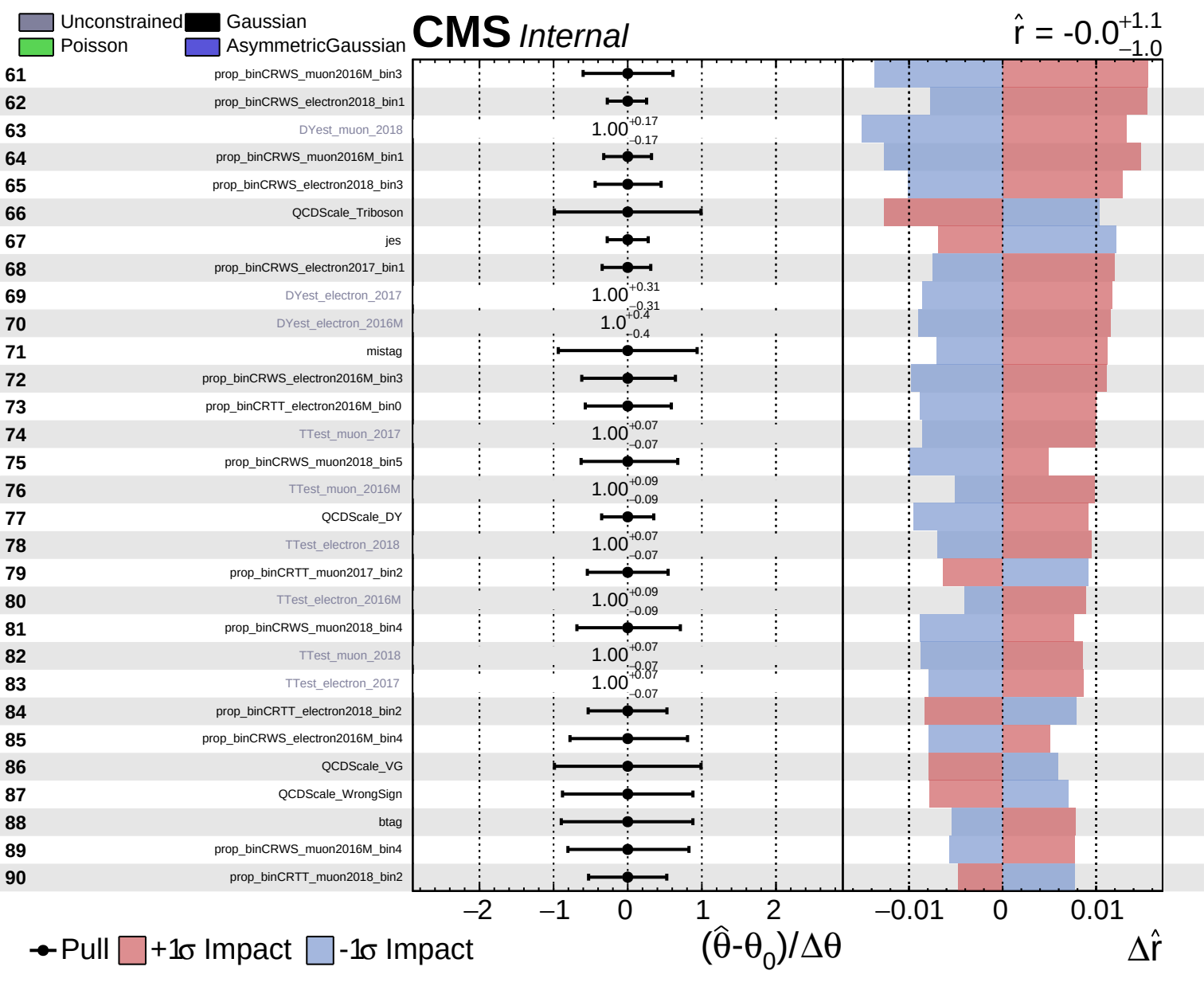
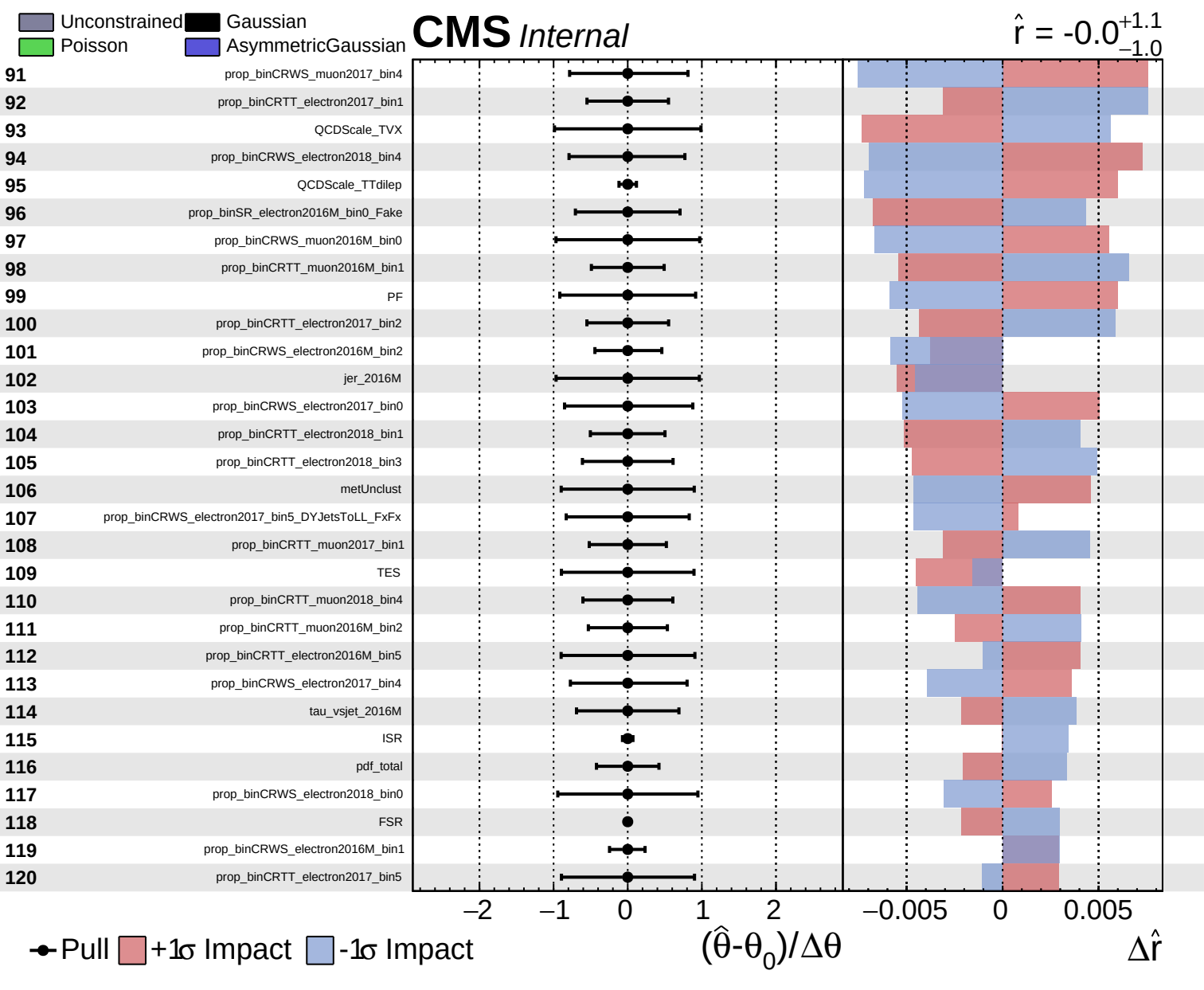
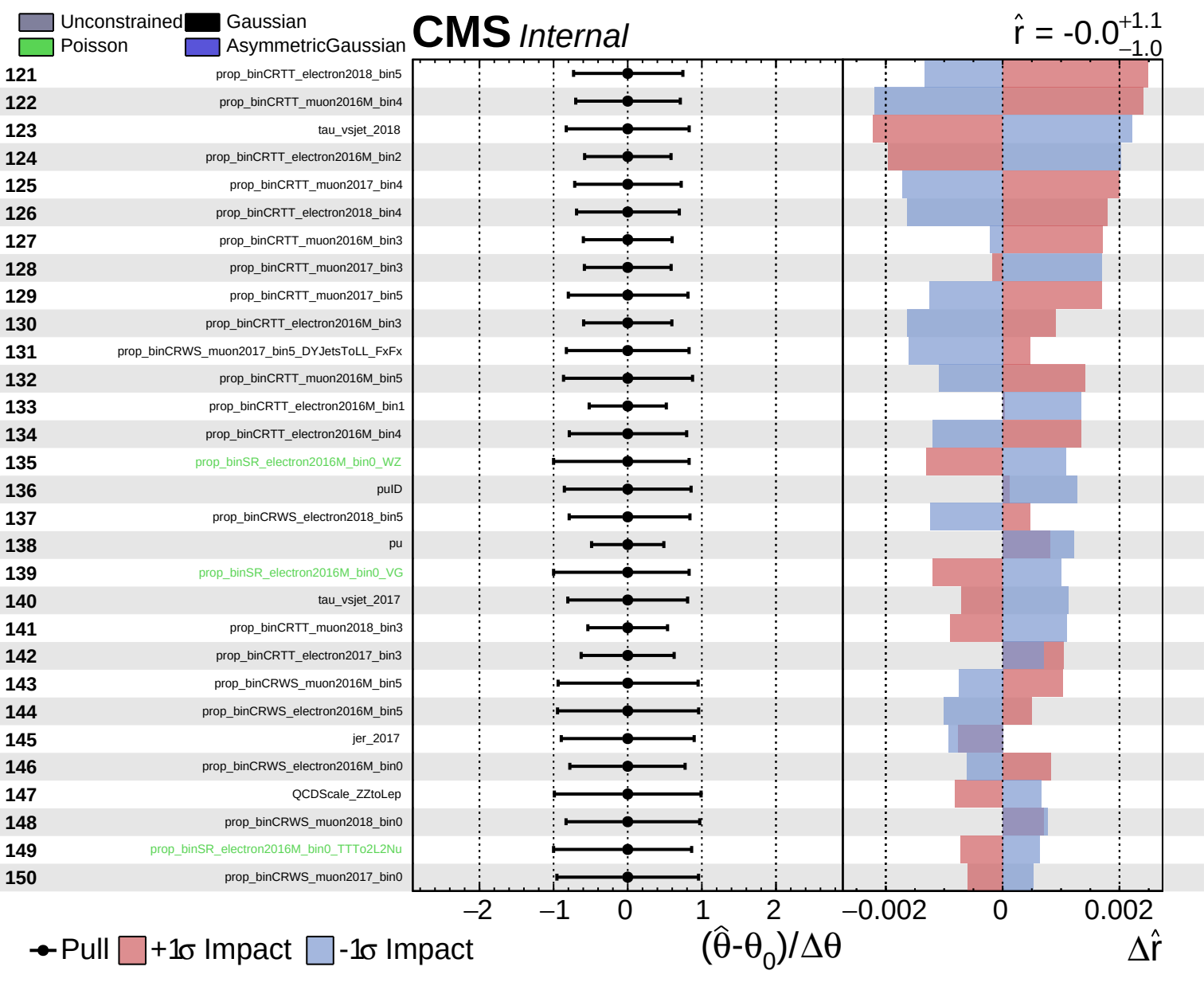
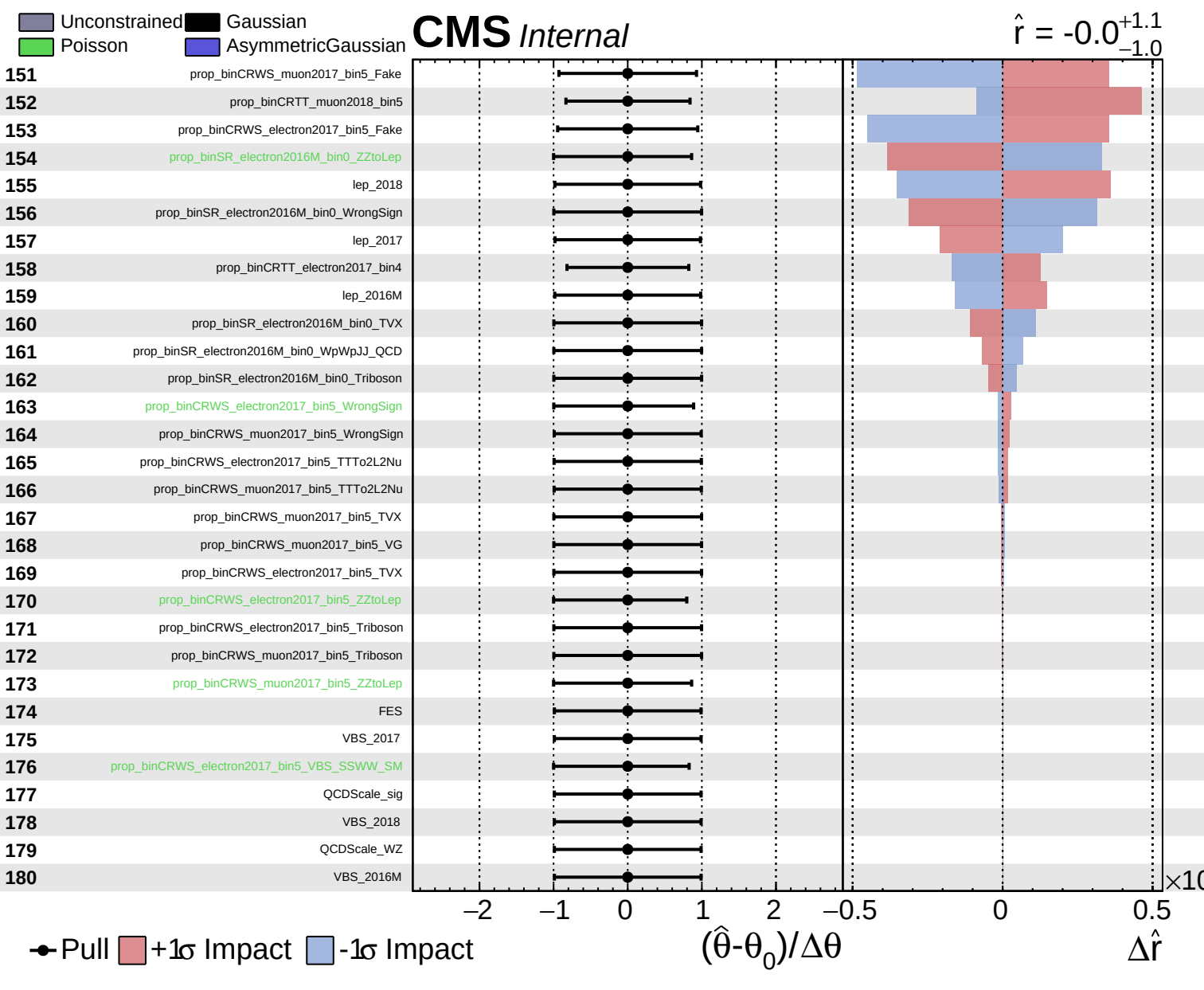

CMS Internal
r̂ = -0.0 +1.1 −1.0
1 prop_binSR_muon2018_bin3
2 prop_binSR_muon2017_bin3
3 prop_binSR_muon2016M_bin3
4 prop_binSR_electron2018_bin3
5 prop_binSR_electron2017_bin3
6 prop_binSR_muon2018_bin2
7 prop_binSR_muon2018_bin4
8 prop_binSR_muon2017_bin2
9 prop_binSR_muon2018_bin5
10 prop_binSR_muon2017_bin4
11 prop_binSR_electron2018_bin4
12 prop_binSR_electron2016M_bin3
13 prop_binSR_muon2016M_bin2
14 prop_binSR_electron2018_bin2
15 prop_binSR_electron2018_bin5
16 prop_binSR_electron2017_bin2
17 prop_binSR_muon2016M_bin4
18 prop_binSR_electron2017_bin4
19 prop_binSR_muon2017_bin5
20 prop_binSR_muon2018_bin1
21 prop_binSR_muon2017_bin1
22 prop_binSR_electron2016M_bin2
23 prop_binSR_muon2018_bin0
24 prop_binSR_muon2017_bin0
25 prop_binSR_muon2016M_bin1
26 prop_binSR_electron2018_bin1
27 prop_binSR_muon2016M_bin5
28 prop_binSR_electron2017_bin5
29 prop_binSR_electron2016M_bin4
30 prop_binSR_muon2016M_bin0
−2 −1 0 1 2
−0.2 0 0.2
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂

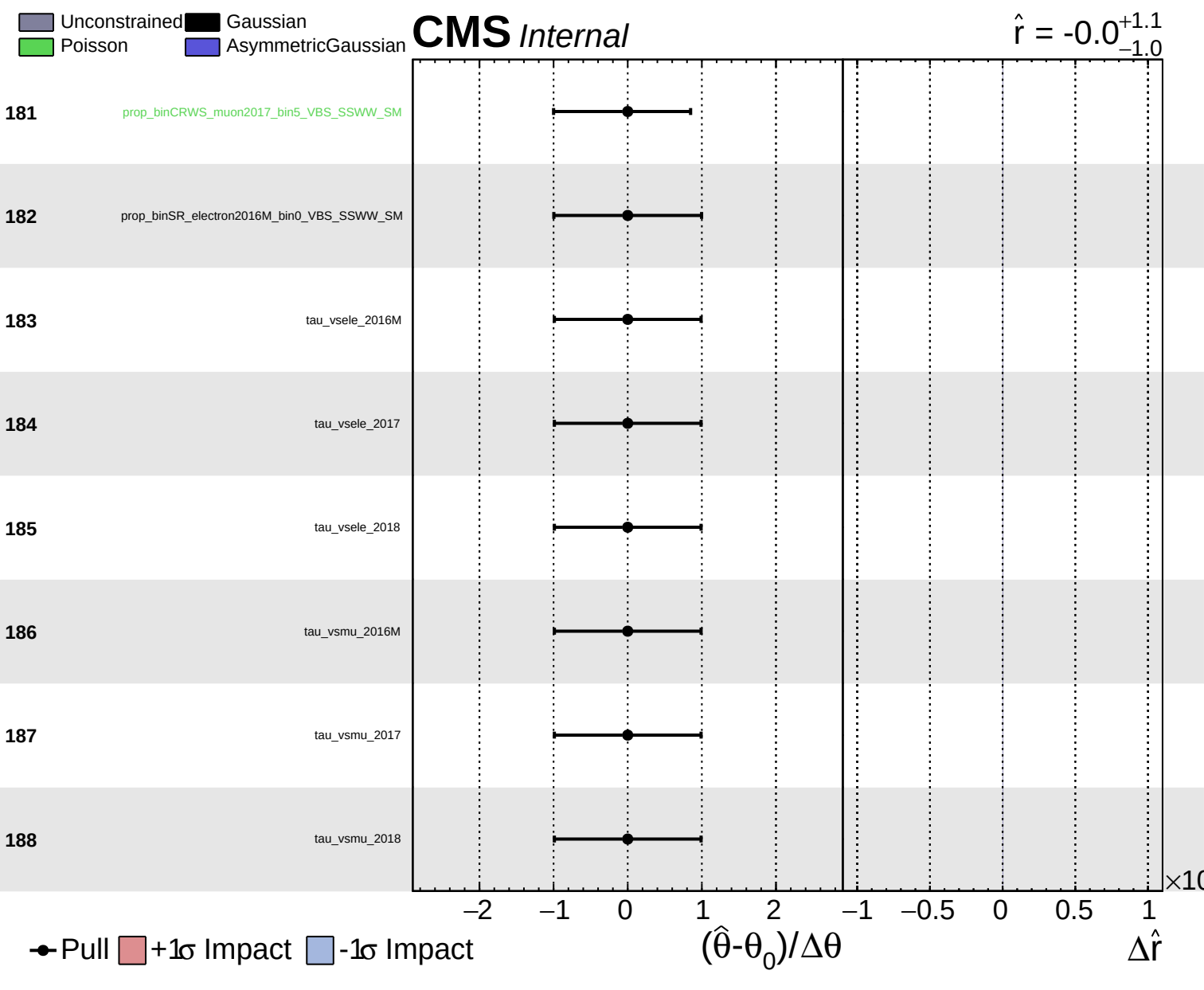

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.0 +1.1 −1.0
31 prop_binSR_electron2017_bin1
32 prop_binSR_electron2016M_bin1
33 prop_binSR_electron2018_bin0
34 prop_binSR_electron2016M_bin5
35 prop_binSR_electron2017_bin0
36 jer_2018
37 lumi_2018
38 lumi_2017
39 lumi_2016M
40 prop_binCRWS_muon2018_bin2
41 prop_binCRWS_muon2017_bin3
42 prop_binCRWS_muon2016M_bin2
43 prop_binCRTT_muon2018_bin0
44 prop_binCRWS_electron2018_bin2
45 prop_binCRWS_electron2017_bin2
46 prop_binCRWS_muon2018_bin1
47 prop_binCRTT_electron2018_bin0
48 prop_binCRWS_muon2017_bin1
49 QCDScale_WpWpJJ_QCD
50 DYest_electron_2018
1.0 +0.4 −0.4
51 prop_binCRWS_muon2018_bin3
52 DYest_muon_2017
1.00 +0.24 −0.24
53 prop_binCRWS_muon2017_bin2
54 prop_binSR_electron2016M_bin0_DYJetsToLL_FxFx
55 prop_binCRTT_muon2018_bin1
56 prop_binCRWS_electron2017_bin3
57 prop_binCRTT_muon2016M_bin0
58 DYest_muon_2016M
1.00 +0.30 −0.30
59 prop_binCRTT_electron2017_bin0
60 prop_binCRTT_muon2017_bin0
−2 −1 0 1 2
−0.05 0 0.05
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.0 +1.1 −1.0
61 prop_binCRWS_muon2016M_bin3
62 prop_binCRWS_electron2018_bin1
63 DYest_muon_2018
64 prop_binCRWS_muon2016M_bin1
65 prop_binCRWS_electron2018_bin3
66 QCDScale_Triboson
67 jes
68 prop_binCRWS_electron2017_bin1
69 DYest_electron_2017
70 DYest_electron_2016M
71 mistag
72 prop_binCRWS_electron2016M_bin3
73 prop_binCRTT_electron2016M_bin0
74 TTest_muon_2017
75 prop_binCRWS_muon2018_bin5
76 TTest_muon_2016M
77 QCDScale_DY
78 TTest_electron_2018
79 prop_binCRTT_muon2017_bin2
80 TTest_electron_2016M
81 prop_binCRWS_muon2018_bin4
82 TTest_muon_2018
83 TTest_electron_2017
84 prop_binCRTT_electron2018_bin2
85 prop_binCRWS_electron2016M_bin4
86 QCDScale_VG
87 QCDScale_WrongSign
88 btag
89 prop_binCRWS_muon2016M_bin4
90 prop_binCRTT_muon2018_bin2
1.00 +0.17 −0.17
1.00 +0.31 −0.31
1.0 +0.4 −0.4
1.00 +0.07 −0.07
1.00 +0.09 −0.09
1.00 +0.07 −0.07
1.00 +0.09 −0.09
1.00 +0.07 −0.07
1.00 +0.07 −0.07
1.00 +0.07 −0.07
−2 −1 0 1 2
−0.01 0 0.01
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.0+1.1−1.0
91 prop_binCRWS_muon2017_bin4
92 prop_binCRTT_electron2017_bin1
93 QCDScale_TVX
94 prop_binCRWS_electron2018_bin4
95 QCDScale_TTdilep
96 prop_binSR_electron2016M_bin0_Fake
97 prop_binCRWS_muon2016M_bin0
98 prop_binCRTT_muon2016M_bin1
99 PF
100 prop_binCRTT_electron2017_bin2
101 prop_binCRWS_electron2016M_bin2
102 jer_2016M
103 prop_binCRWS_electron2017_bin0
104 prop_binCRTT_electron2018_bin1
105 prop_binCRTT_electron2018_bin3
106 metUnclust
107 prop_binCRWS_electron2017_bin5_DYJetsToLL_FxFx
108 prop_binCRTT_muon2017_bin1
109 TES
110 prop_binCRTT_muon2018_bin4
111 prop_binCRTT_muon2016M_bin2
112 prop_binCRTT_electron2016M_bin5
113 prop_binCRWS_electron2017_bin4
114 tau_vsjet_2016M
115 ISR
116 pdf_total
117 prop_binCRWS_electron2018_bin0
118 FSR
119 prop_binCRWS_electron2016M_bin1
120 prop_binCRTT_electron2017_bin5
−2 −1 0 1 2
−0.005 0 0.005
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.0+1.1−1.0
121 prop_binCRTT_electron2018_bin5
122 prop_binCRTT_muon2016M_bin4
123 tau_vsjet_2018
124 prop_binCRTT_electron2016M_bin2
125 prop_binCRTT_muon2017_bin4
126 prop_binCRTT_electron2018_bin4
127 prop_binCRTT_muon2016M_bin3
128 prop_binCRTT_muon2017_bin3
129 prop_binCRTT_muon2017_bin5
130 prop_binCRTT_electron2016M_bin3
131 prop_binCRWS_muon2017_bin5_DYJetsToLL_FxFx
132 prop_binCRTT_muon2016M_bin5
133 prop_binCRTT_electron2016M_bin1
134 prop_binCRTT_electron2016M_bin4
135 prop_binSR_electron2016M_bin0_WZ
136 puID
137 prop_binCRWS_electron2018_bin5
138 pu
139 prop_binSR_electron2016M_bin0_VG
140 tau_vsjet_2017
141 prop_binCRTT_muon2018_bin3
142 prop_binCRTT_electron2017_bin3
143 prop_binCRWS_muon2016M_bin5
144 prop_binCRWS_electron2016M_bin5
145 jer_2017
146 prop_binCRWS_electron2016M_bin0
147 QCDScale_ZZtoLep
148 prop_binCRWS_muon2018_bin0
149 prop_binSR_electron2016M_bin0_TTTo2L2Nu
150 prop_binCRWS_muon2017_bin0
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.0+1.1−1.0
151 prop_binCRWS_muon2017_bin5_Fake
152 prop_binCRTT_muon2018_bin5
153 prop_binCRWS_electron2017_bin5_Fake
154 prop_binSR_electron2016M_bin0_ZZtoLep
155 lep_2018
156 prop_binSR_electron2016M_bin0_WrongSign
157 lep_2017
158 prop_binCRTT_electron2017_bin4
159 lep_2016M
160 prop_binSR_electron2016M_bin0_TVX
161 prop_binSR_electron2016M_bin0_WpWpJJ_QCD
162 prop_binSR_electron2016M_bin0_Triboson
163 prop_binCRWS_electron2017_bin5_WrongSign
164 prop_binCRWS_muon2017_bin5_WrongSign
165 prop_binCRWS_electron2017_bin5_TTTo2L2Nu
166 prop_binCRWS_muon2017_bin5_TTTo2L2Nu
167 prop_binCRWS_muon2017_bin5_TVX
168 prop_binCRWS_muon2017_bin5_VG
169 prop_binCRWS_electron2017_bin5_TVX
170 prop_binCRWS_electron2017_bin5_ZZtoLep
171 prop_binCRWS_electron2017_bin5_Triboson
172 prop_binCRWS_muon2017_bin5_Triboson
173 prop_binCRWS_muon2017_bin5_ZZtoLep
174 FES
175 VBS_2017
176 prop_binCRWS_electron2017_bin5_VBS_SSWW_SM
177 QCDScale_sig
178 VBS_2018
179 QCDScale_WZ
180 VBS_2016M
−2
−1
0
1
2
−0.5
0
0.5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.0+1.1−1.0
181
prop_binCRWS_muon2017_bin5_VBS_SSWW_SM
182
prop_binSR_electron2016M_bin0_VBS_SSWW_SM
183
tau_vsele_2016M
184
tau_vsele_2017
185
tau_vsele_2018
186
tau_vsmu_2016M
187
tau_vsmu_2017
188
tau_vsmu_2018
×10
−2
−1
0
1
2
−1
−0.5
0
0.5
1
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂